



Methods for Measuring Sub-surface Skin Displacement During Ultrasound Mid-air Haptic Stimulation

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Abstract. Ultrasound mid-air haptic interfaces allow for the rendering of tactile stimulation with no contact, through the constructive interference of phase-controlled acoustic waves. In addition to the interesting potential it brings for rich and natural tactile stimulation, its large design space and noncontact nature stand out from most existing technologies. This means that UMH requires more in-depth understanding, especially of the skin mechanics in response to stimulation, and how it contributes to tactile perception. At the same time, the noncontact nature of UMH facilitates such study, as it can stimulate the skin from a distance, without occluding measurement tools. Here, we propose an original setup to measure the sub-surface displacement of the glabrous skin in response to a variety of UMH stimuli, using functional optical coherence tomography (fOCT). Such measures are necessary to understand how the skin reacts to the acoustic radiation of UMH, providing insights about the underlying perception of UMH stimuli. We present the first set of measurements combining UMH and fOCT with a data processing pipeline and preliminary discussion of early results demonstrating feasibility.

Keywords: Ultrasound mid-air haptics · Optical coherence tomography · Sub-surface skin imaging

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1 Introduction and Related Work

Most tactile stimulators rely on the application of forces on the skin, leading to small indentations and displacements, usually representing information to the recipient (e.g. texture [1]). Once the initial force has been applied, the mechanical energy propagates, both on the surface and below [9, 24]. If the resulting tissue displacement is large enough when reaching a sub-surface receptor ending, it may trigger an action potential, which is the first physiological step initiating the perception process [12]. This highlights the crucial role of the skin behaviour in the transmission and perception of tactile stimuli.

When using a tactile interface, we usually have control over the applied stimulus (e.g. intensity of a vibration), but the skin's response is rarely taken into account during the rendering process. Yet, a better understanding of the skin's behaviour could greatly improve our understanding of the rendering-perception interaction, which is necessary for the design of efficient rendering algorithms. Some studies have therefore attempted to measure the displacements applied to the skin, through different means. A common method relies on the use of laser Doppler vibrometers (LDV) to assess the speeds and accelerations of displaced tissues at the surface of the skin (e.g. [5, 8]). Yet these methods are usually limited to measuring displacements at the surface of the skin, while tactile afferents are actually located sub-surface. Recently developed approaches rely on optical coherence tomography (OCT). Such systems use interference measurements of a light beam targeted at the skin to acquire properties of the medium at different depth levels, up to a few millimeters in total depth. OCT techniques, in addition to producing structural images (structural OCT), are able to capture tissue motion at a high sampling rate (functional OCT), and have recently been used to analyze skin displacement in response to contact vibrotactile stimulation [10], brushing [25] and quasistatic contact [9]. These approaches can analyze tissue displacement by measuring the phase variation of the signal [30], and measuring its variation in the frequency domain [4, 31].

On the other hand, ultrasound mid-air haptic technologies rely on the interference of phase-controlled acoustic waves to generate focal points of high acoustic intensity in mid-air [23]. The amplitude or position of this focal point can then be modulated through processes referred to as amplitude modulation (AM) [16, 17] and spatiotemporal modulation (STM) [11, 19] respectively, to provide low-intensity noncontact tactile stimulation. This process of rendering tactile stimuli relies on acoustic radiation pressure, which is fundamentally different from most other types of tactile interfaces relying on mechanical contact. Consequently, the skin's mechanical response to these low-intensity, non-contact stimuli likely differs significantly from its response to contact-based inputs. Yet, skin response to UMH stimulation has very much been understudied, with only a few studies using LDV to measure surface indentation [6, 8, 11, 24], but none directly imaging sub-surface displacements. However, such insights could be crucial to optimize UMH stimulus rendering methods in order to increase the perceived intensity, which is currently one of the main limitations of UMH. This paper aims to address this gap. Interestingly, the noncontact nature of UMH makes

it well suited for such study, as the interface can be moved outside the field of view of the OCT, preventing any occlusions that would otherwise happen with contact haptic devices.

Here, we make the following contributions: 1) a setup for measuring the sub-surface skin displacement induced by UMH stimulation, based on OCT (Sect. 2); 2) the first proof-of-concept OCT measurements during UMH stimulation, serving to validate the apparatus and highlighting the feasibility of using our framework to provide preliminary insights into skin displacement dynamics for different UMH rendering schemes (Sect. 3).

2 The Setup: Combining Ultrasound Mid-air Haptics and Optical Coherence Tomography

To measure sub-surface skin displacement in response to UMH stimulation, we designed a setup combining both OCT technologies for data acquisition, and a UMH interface for tactile stimulation. The overall setup is illustrated in Fig. 1.

Hardware. We used a spectral domain optical coherence tomography (SD-OCT) [29] imaging system [TEL320C1, Thorlabs, USA] to capture sub-surface skin displacements. This system uses superluminescent diodes to produce broadband light focused into a $13\text{ }\mu\text{m}$ diameter beam, then split into a reference and a sample arm. The sample arm light penetrates the skin, reflects back, and recombines with the reference arm light. The combined signal is then captured by a spectrometer consisting of a grating and a 2,048-pixel 1D CCD camera. The broadband nature of the light source allows the captured spectrum to be transformed into intensity and phase values (represented as complex numbers) at 1,024 depth samples, creating a 1D A-scan or A-line image from the raw CCD camera data. Specifically, the amplitude of the complex number represents the amount of light reflected by the internal structures at the given depth level, while the phase informs about their relative position within each pixel. Specifically, with this hardware, each pixel corresponds to around $3.5\text{ }\mu\text{m}$ in the air, or $2.5\text{ }\mu\text{m}$ in the skin. For this study, the beam was held at a fixed location to prioritize temporal resolution, enabling high-speed recording of skin motion at 10 kHz. This method of staying at one location and recording overtime is called an M-scan (or M-mode) [4]. The OCT was attached to a collaborative robot arm [UR3e, Universal Robots, Denmark], allowing for precise translation and orientation using its native controller, in order to focus the imaging system to the desired area on the skin.

For the tactile stimulation, we relied on a custom UMH interface, composed of a 16×16 array of transducers generating a 40 kHz carrier. Focal point positions and intensities were updated at 40 kHz. It was attached to a moveable support arm [Mini Salon 806, Manfrotto, Italy], ensuring easy manipulation of the interface to target the desired stimulation location. The interface was positioned 8 cm away from the skin, and stimuli were generated at the surface of the skin. With our specific interface, this focal distance maximizes the acoustic

radiation pressure generated at the skin surface. According to the results of [16], our device generates a pressure distribution with a diameter of 8.2 mm at 8 cm distance.

During recording sessions, the participant was laying in a Fowler's position on a treatment chair, with cushions for comfort. Their right arm was positioned on an armrest, and stabilized using a vacuum-pump cushion, as illustrated in Fig. 1.

Software. The OCT and UMH systems were triggered by separate computers (PC1 and PC2). To synchronize the recording with the tactile stimulation, the TTL signal sent from PC1 to the OCT to trigger the recording was detected by a development board [Uno R3, Arduino, Italy]. The board then informed PC2 of the upcoming recording through serial connection, allowing PC2 to trigger the tactile stimulation.

Custom handlers were used to control each of the two systems. A Python wrapper around a LabVIEW control system was used for communication with the OCT, while a custom Python wrapper around a C++ experiment program was used for producing the UMH stimulation. Interactions between all components are illustrated in Fig. 1.

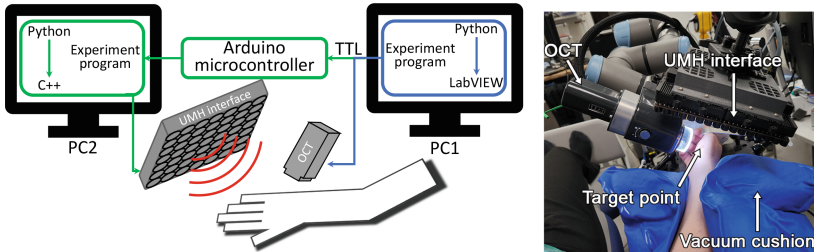


Fig. 1. (Top) General setup for OCT recording during UMH stimulation. (Blue) The driving computer sends a TTL signal to trigger the acquisition of each A-line. (Green) The first TTL is also detected by an Arduino microcontroller, which informs the second computer, leading to the rendering of the next haptic condition. (Bottom) Experimental setup for recording on the palm: the participant's arm is stabilized using a vacuum cushion, while both the OCT and UMH interface are targeted at the same location on the skin.

Discussion on the Setup. While the proposed setup provides interesting and innovative potential, the combination of OCT and UMH technologies imposes some physical limitations. First of all, the OCT needs to be placed very close to the skin in order to provide quality measurements. This necessitates placing the OCT within the acoustic propagation path, creating a partial physical obstruction. During stimulation, the OCT could thus be absorbing or reflecting

some ultrasound waves. This may lead to forces applied at the stimulation that are lower than expected, reducing the stimulus intensity. We empirically verified that any absorption was not sufficient to move or vibrate the OCT. Therefore, it did not generate additional noise in the recorded signal.

To reduce such absorption effect, both the OCT and UMH interface could be tilted to image and stimulate the skin at an angle, instead of the usual normal stimulation. Yet, such diagonal setup still somewhat reduces the radiation force applied to the skin, modelled with the following equation: $F = PS \cos \theta$, where F [N] is the radiation force, P [Pa] is the acoustic radiation pressure generated at the target location, S [m²] is the surface area undergoing acoustic radiation, and θ is the angle between the skin's normal direction and the UMH interface's normal direction [3, 15]. As the OCT's light source travels in a straight line, tilting the OCT means that the obtained A-lines are acquired at a diagonal to the skin. This reduces the effective depth at which we can measure skin displacement. For this reason, all reported depths in this article are expressed as distances along the path of the light beam. Such effects may be mitigated in the future by using acoustic reflectors [2] to enforce a more optimal route for the acoustic wave that would allow normal OCT positioning and quasi normal acoustic incidence.

Recording Study: Sub-surface Glabrous Skin Response to Ultrasound Mid-air Haptic Stimulation. We then leveraged our setup to perform the first sub-surface skin displacement measurements in response to UMH stimulation, aiming to validate our system and provide preliminary discussions of the impact of UMH rendering methods on skin displacement.

We delivered a total of 20 stimuli, corresponding to the combination of the following factors: 1) **rendering technique**, with two levels: either a static point rendered using AM, or a STM circle with 1 cm radius, both centered on the OCT measurement point; 2) **modulation frequency**, with 10 levels: 5, 10, 30, 50, 100, 200, 250, 300, 400, or 500 Hz. STM focal speeds were calculated based on the modulation frequency and trajectory diameter, ranging from 0.16 m/s to 16 m/s. These conditions were chosen as they correspond to the simplest and most common stimulation patterns with UMH [11, 14, 16, 19], with frequencies ranging from low-frequency generating non-vibratory static pressure sensations [20, 21] to very high frequencies near the upper limit for tested vibrotactile perception [18]. We also used a denser frequency sampling near the 200–300 Hz frequency, as this frequency range is often used for UMH stimulation due to the higher perceived intensity provided [13, 14], possibly due to the selective activation of afferents innervating Pacinian corpuscles [18]. It is worth noting that we technically recorded from more types of stimuli, as well as their fingertip response, but only discuss the AM point and STM circle on the palm here for the sake of validating our framework. Deeper analyses will be conducted on this wider range of stimuli in the future.

Each stimulus was rendered for a duration of 200 ms, while the OCT recorded for a total duration of 500 ms with a 10 kHz sample rate, ensuring that the pre- and post-stimulation behaviour are also captured (see Fig. 2). Stimuli were ren-

dered following a constant order: all AM stimuli are rendered successively with increasing modulation frequency, before all STM stimuli are rendered similarly. Since the OCT recording window is longer than the UMH stimulation duration, there is a stimulation pause of 300 ms between stimuli, which allows for skin relaxation. Recordings were conducted on the glabrous skin of the participant’s right hand, on the thenar eminence.

3 Results and Proof of Feasibility

Here, we present the analysis pipeline to analyze the skin’s response to UMH stimulation, based on the recording experiment presented in Sect. 2. This provides insights for deeper futur analysis, and a proof of feasibility.

Collected Data. For each of the stimuli described in Sect. 2, we collected a M-scan of 500 ms with a sampling rate of 10 kHz at the same skin location. This resulted in 20 M-scans corresponding to glabrous skin on the thenar eminence of a single participant.

Analysis and Discussion. No movement was qualitatively observed on amplitude data, indicating that UMH motion amplitudes were sub OCT pixel (below approximatively $2.5\text{ }\mu\text{m}$). This renders standard OCT intensity data insufficient. Instead, we use the phase to track sub-resolution displacement. While studies report sub-nanometer sensitivity [7], our system is known to track motion down to 100 nm [31], which is well within the UMH range. The phase data from each M-scan was rendered as a 2D image, where the y-axis represents depth, and the x-axis represents time. To isolate displacement from phase constant, we computed the differential phase between consecutive A-scan (columns), which provides tissue displacement information [30] as illustrated in Fig. 2. This is the input data for our analysis pipeline, where we perform frequency domain analyses [4, 31]. Already, we observe vertical lines of rapid phase change, corresponding to each modulation period.

We manually selected a row r_S corresponding to the surface of the skin. Each row below r_S can thus be interpreted as a 1D time series, corresponding to the phase differential signal at a given depth in the skin. We performed a continuous wavelet transform (CWT) on these time series, using a complex Morlet wavelet (bandwidth = 1.5, center frequency = 1) on 128 linearly spaced frequencies between 1 and 1000 Hz. The $g(W(f, t)) = |W(f, t)|^2$ transformation was applied to each of the obtained $W(f, t)$ coefficients, corresponding to each frequency f and time step t . This resulted in a real-valued matrix of time-frequency “power” coefficients. Then, all power coefficients corresponding to time steps before or after stimulation were discarded to remove potential noise. Similarly some low-frequency noise was always recorded, in part due to physiological phenomena (e.g. blood flow, breathing). We therefore discarded all power coefficients corresponding to frequencies lower than 75 Hz (frequency chosen empirically based

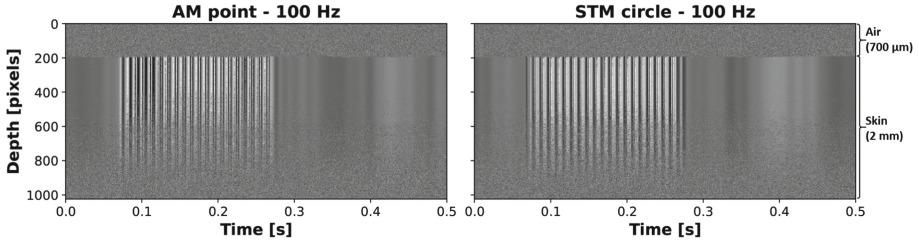


Fig. 2. Phase slice data for the 100 Hz stimuli rendered with AM (left) and STM (right). We observe vertical lines of rapid change during stimulation. The number of lines correlates with the modulation frequency. Brighter pixels represent motion towards the probe. Pixels above the skin surface ($y \approx 200$) correspond to $3.5 \mu\text{m}$, while skin pixels correspond to approximately $2.5 \mu\text{m}$.

on the observed noise bandwidth). This also led us to discard conditions using modulation frequencies below this cutoff frequency, as the stimulation frequency would be indistinguishable from the noise. A different analysis pipeline is needed to analyse low frequencies, and is beyond the scope of this article. Examples of the resulting power matrices are shown in Fig. 3.

For each power matrix, we computed the time-averaged power at every wavelet frequency, providing insights about how the UMH energy gets distributed across different wave frequencies. By plotting this distribution (see Fig. 3), we can usually observe one to three local maxima, around the modulation frequency and the next two harmonics. These can also be visualized in the previous power matrices, corresponding to horizontal bands of higher power. Overall, this shows that our system can successfully detect the stimulus, as well as its modulation frequency and first harmonics.

Next, on each of the power matrices obtained (meaning for each combination of stimulus type, modulation frequency, and depth), we computed the time-averaged power in a 50 Hz frequency window centered around the corresponding modulation frequency (value chosen empirically: it is enough to collect the energy from most of the band, without interfering with the other harmonics). This metric serves as a representation of the amount of energy directly converted from the acoustic energy of the stimulation, to a same-frequency wave in the skin. While Fig. 3 showed that higher-frequency were involved, it also highlights that most of the energy usually corresponds to displacements at frequencies near the modulation frequency. By plotting the depth evolution of this metric (see Fig. 4), we can visualize the energy decay. The observed decay rate seems highly non-constant, but similar profiles appear between AM and STM, especially at frequencies from 300 Hz. While more data is required to confirm the observed behaviours, different regions seem to appear: 1) from the surface of the skin to 0.5–0.75 mm depth, the amount of energy seems to be mostly constant, or even slightly increasing with depth; 2) we then see a mostly linear decrease in energy until the 1–1.25 mm depth; 3) we then observe a second plateau, followed with a final decay. It is possible that these observed regions correspond to different parts

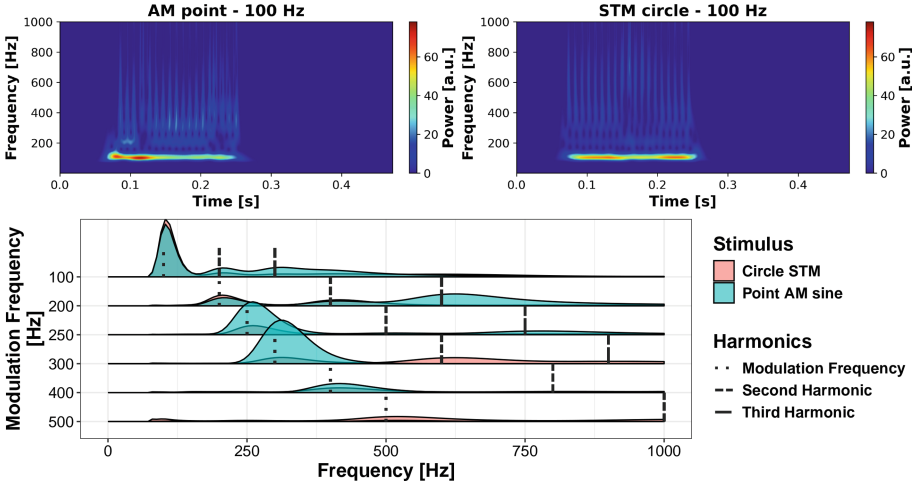


Fig. 3. (Top) Filtered CWT of the OCT data corresponding to the surface of the skin with the 100 Hz AM (left) and STM (right) stimuli. Horizontal lines of high energy can be observed at the modulation frequency and (more faint) at the next two harmonics. Vertical lines also appear every modulation period. (Bottom) Time-averaged power distribution across frequencies, computed from the associated CWTs. Local maxima appear around the modulation frequency and following harmonics. Such figures can be computed at each pixel depth.

of the skin (e.g. region #1 may correspond to the skin’s stratum corneum), with plateaus being caused by between-layer boundaries. Future efforts could cross-reference these transitions with skin layer boundaries using structural images.

Overall, Fig. 3 hints towards differences between the skin’s response to AM and STM. Notably, AM seems to consistently provide a similar or higher-amplitude response from the skin at the surface. While this could first seem counter-intuitive, due to the usually higher perceived intensity of STM (e.g. [22]) and higher delivered energy overall [19], a few reasons may explain such results. First, the STM circle’s center, AM point, and target location for the A-line were co-localized. This means that, for the STM conditions, the focal point is moving around the A-line, remaining at a 1 cm distance. The measured response therefore corresponds to waves that have travelled from the focal point to the recording location. Yet due to the viscoelasticity of the skin, the energy conveyed by these waves decays with lateral distance, resulting in lower recorded power. Second, while STM generates more energy overall, this energy is distributed across the whole rendered pattern (here, a circle). So each individual location on the scanning trajectory actually receives less energy than AM [19]. The higher perceived intensity may therefore be due to spatial summation effects. Future work will be focused on reproducing the experiment with more repetitions and participants to assess such effects in more details.

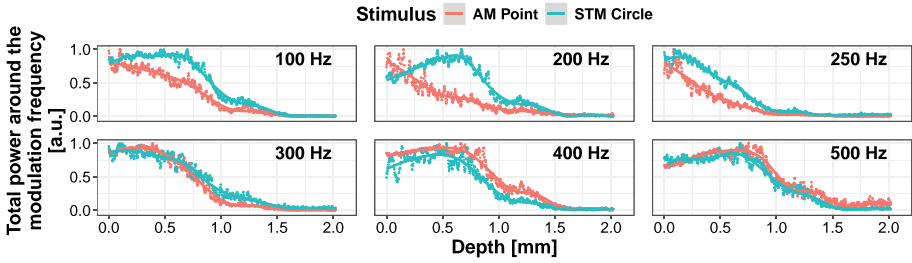


Fig. 4. Depth variation of the CWT power concentrated in a 50 Hz window around the modulation frequency. Depth = 0 corresponds to the surface of the skin. Each subplot corresponds to its own modulation frequency, and every signal was rescaled to $[0, 1]$. Cubic splines were fitted for each stimulus.

Interestingly, the width of the power distribution around the main harmonics seem to increase with the modulation frequency, but no such effect was observed when using a Fourier transform instead of a CWT. While this correlates with the frequency-dependant response of the skin [32,33], future investigations should focus on formally comparing the results with existing frameworks for the skin response (e.g. [26–28]), and recruit more participants to better quantify such potential effects.

Limitations and Perspectives. The above results and discussion present a good proof of concept of our setup and pipeline, showing good potential for deeper analyzing the skin’s response to UMH stimulation. Yet, a few current limitations need to be addressed. First, we currently recorded on a single participant, yielding limited amount of data. While we already observe some trends, a higher amount of participants and repetitions would be necessary to provide more confident and generalized conclusions. This could clarify the current small inconsistencies we observe between modulation frequencies and depth.

Secondly, as mentioned in Sect. 2, combining OCT with UMH currently requires tilting both of them, which leads to lower radiation forces being applied to the skin, and diagonal A-lines. This implies that the amplitude of actual effects would be stronger in a more usual interaction with a UMH interface. The diagonal recording also means that we cannot record as deep in the skin as we would with a perpendicular recording setup. For reference, we expect that the effective depth of the measurements was reduced by approximately 13–50%, compared to what would be possible using a perpendicular measurement. Future work should look into ways to redirect the acoustic field to allow for stronger stimulation and more upright positioning of the OCT.

Finally, while we provided a proof of feasibility to analyse the obtained data for frequencies above 75 Hz, we had to discard low-frequency noisy data. Future work will focus on optimizing our setup to reduce such noise, and adapt our pipeline to analyse low-frequency effects.

4 Conclusion

To address the current lack of knowledge regarding the skin's response to mid-air haptic technologies, we proposed a setup combining functional optical coherence tomography (fOCT) and ultrasound mid-air haptics (UMH) to visualize sub-surface displacement of the skin during stimulation. We used this setup to record the palm's skin response to UMH stimuli rendered with different modulation methods and frequency. This provides proof of feasibility in observing such response. Preliminary results hint towards a non-trivial decay with depth, as well as the concentration of energy around the modulation frequency and its harmonics. Future work will focus on acquiring more data on more participants to confirm, generalize, and better quantify the currently observed trends. We will also further improve the setup using acoustic reflectors to allow for normal incidence and thus higher radiation pressure.

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